
VoteVox – Administrator Manual

QR-code based digital voting system for HTL Wels

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1. About this manual

This manual is for **administrators** (e.g. teachers or the election committee) who prepare and run an election with VoteVox.

It has two parts:

- **Part A – Test program:** describes the system exactly as it runs today (diploma project / demo) on a single PC.
- **Part B – Production version:** describes what the system would look like in a later, productive deployment at the school.

Terms - Admin UI: the management interface (administrators only). - **Voting UI:** the voting interface students use on their phones. - **Voting code / token:** a one-time, anonymous code printed as a QR code that allows exactly one vote to be cast.

2. System overview

VoteVox consists of four building blocks that work together automatically on start-up:

Building block	Purpose	Address (test program)
Database	Stores elections, classes, codes and votes	(internal)
Backend	The "logic": validates codes, counts votes	(internal, port 8080)
Admin UI	Management of elections	<code>https://localhost:5174</code>
Voting UI	Vote casting by students	<code>https://<PC-IP>:5173/vote</code>

Core principle – anonymity: No student names are stored. Per participating class, only a number of anonymous voting codes is generated. The system does not know which code belongs to which person – only the class is recorded. Each code can be used **exactly once**.

Part A – Test program (current version)

A1. Start the system

1. Make sure **Docker Desktop** is installed.
2. Double-click **VoteVox starten.bat** in the project folder.
3. A window opens that automatically:
 - checks whether Docker is running (and starts Docker if needed),
 - detects the PC's current network address (IP),
 - creates an HTTPS certificate for that address,
 - builds and starts all building blocks,
 - opens the admin interface in the browser.

On the very first start, Docker downloads many components and builds the system – this can take a few minutes. After that it is much faster.

At the end the window shows a summary with all important addresses, including the voting address for the phones (e.g. `https://192.168.178.44:5173/vote`). This window can be closed – VoteVox keeps running in the background.

 **Important:** If Windows asks for a **firewall exception** on the first start, it must be **allowed**, otherwise phones cannot reach the PC.

A2. Sign in

1. The admin interface opens at `https://localhost:5174` .
2. Default credentials in the test program:
 - **E-mail:** `admin@votevox.at`
 - **Password:** `Admin1234!`
3. After signing in you land on the **Dashboard** with an overview.

If the browser shows a certificate warning: there should be none on the PC. For phones see [A7](#).

A3. Manage classes

Before an election can be run, the participating **classes** must exist.

1. Open the **“Classes”** menu item.
2. Use **“Add class”** to create a class:

- **Name** of the class (e.g. 4AHIT),
 - **Student count**.
3. The count determines how many anonymous voting codes are generated for that class.

The test program already contains three sample classes (e.g. 4AHIT , 4BHIT , 4AHWI).

A4. Create an election

1. “**Elections**” → “**New election**”.
2. Fill in the fields:
 - **Title** (e.g. “School Representative Election 2026”),
 - **Type** of election,
 - **Description** (optional),
 - **Start date** and **End date**.
3. Save. The election now has the status “**Planned**”.

An election moves through the states **Planned** → **Running** → (**Stopped/Closed**). Voting codes can only be generated while the status is **Planned**.

A5. Add candidates

1. Open the election and switch to the “**Candidates**” tab.
2. For each candidate enter:
 - **Name**,
 - **Class**,
 - **Department**.
3. You can add any number of candidates (while the election is **Planned**).

A6. Select participating classes

1. In the election switch to the “**Classes**” tab.
2. Tick the classes that may take part in this election (use “**Select all**” to select all at once).
3. The sum of the student counts of all ticked classes is the number of **eligible votes** – and therefore the number of codes to be generated.

A7. Generate and print voting codes (QR codes)

1. In the election switch to the “**Tokens**” tab.
2. Click “**Generate Tokens**” and confirm.

3. The system creates as many anonymous codes per participating class as there are students.
4. The codes are shown in plain text **only once**.
5. **Immediately** click **“Generate PDF with QR Codes”** and save/print the PDF. The PDF contains 12 QR codes per A4 page (3×4) with cut lines, **grouped by class** (one class per page).
6. Cut the printed sheet along the cut lines – each card is one QR code for one vote.

 **Important:** The plain-text codes are stored only **hashed (encrypted)** in the database and **cannot be recovered** afterwards. If you do not save the PDF, you must reset and regenerate the codes.

Lost the PDF? As long as the election is still *Planned*, the Tokens tab offers **“Reset & re-generate codes”**. This invalidates the old codes and creates a completely new set. Previously printed codes will then stop working.

QR codes on phones without a warning (once per device)

So that scanning shows **no** “connection not secure” warning and the **camera** works, the root certificate should be installed once:

1. Copy `votevox-rootCA.pem` from the project folder to the phone (e.g. by e-mail or USB).
2. Install it on the phone (Android: *Settings* → *Security* → *Install a certificate*; iOS: install the profile **and** enable it under *Settings* → *General* → *About* → *Certificate Trust Settings*).

Even without this installation voting usually works – the phone then shows a warning that must be accepted.

(If the launcher had to fall back to a self-signed certificate, there is no `votevox-rootCA.pem` to install – simply accept the one-time warning.)

A8. Start, monitor and stop the election

- **Start:** in the election, top right, **“Start Election”**. From now on votes can be cast. Codes can **no longer** be generated or reset.
- **Monitor:** the **“Overview”** tab continuously shows the number of **votes cast** versus eligible voters.
- **Stop:** **“Stop Election”**. After this no further votes are possible.

A **running** election cannot be deleted. Stop it first, then delete if needed.

A9. View results

1. In the election switch to the **“Results”** tab.
2. There you can see:
 - **Votes per candidate** (with a chart),

- **Turnout per class** (how many codes were issued/used).

Because no names are stored, it is not possible to tell **who** voted how – only **how many** votes each candidate received.

A10. Audit log

The **“Audit”** menu item shows a log of the important actions (e.g. election started/stopped, codes generated). This serves the traceability of the election process – without any link to individual voters.

A11. Stop the system

- Double-click **VoteVox stoppen.bat** . This stops all building blocks.
- The **data is preserved** – on the next start everything is back.
- To start over with an **empty** database, run the stop script with the **-Wipe** option (deletes **all** data irreversibly).

A12. Sending the project to someone else

The recipient builds everything from source with Docker, so you only send the source – **not** the generated folders.

1. Double-click **Projekt-packen.bat** . It creates a clean **VoteVox-<date>.zip** (without **node_modules** , build output, certificates, **.git**) in the folder above the project.
2. Send that single **.zip** .
3. The recipient:
 - installs **Docker Desktop** (and starts it once),
 - unzips the file,
 - double-clicks **VoteVox starten.bat** .

Everything else (network address, certificate, build) happens automatically on the recipient’s machine.

A13. Troubleshooting

Problem	Cause / fix
“Docker is not running”	Start Docker Desktop and wait until it says “Running”; start the script again.
Phone does not open the voting page	The phone must be on the same WiFi as the PC. Allow the Windows firewall exception for ports 5173/5174.

Problem	Cause / fix
Certificate warning on the phone	Install <code>votevox-rootCA.pem</code> on the phone (see A7), or accept the one-time warning.
QR codes point to the wrong/old address	Did the PC's IP change? Just run <code>VoteVox stoppen.bat</code> then <code>VoteVox starten.bat</code> – the new IP is written into the QR codes automatically. Then regenerate and reprint the codes.
"Generate Tokens" is greyed out	The election is not <i>Planned</i> , or no class is selected.
Code rejected when voting	The code was already used, belongs to a different election, or the election is not running (any more).

Part B – Production version (outlook)

This part describes what VoteVox would look like in a **real, permanent deployment** at the school. These points are **not yet** part of the test program but the planned next step.

B1. Central server installation instead of a single PC

Instead of running on a teacher's laptop, VoteVox runs on a **central school server** (or a hosted server). Advantages:

- Reachable via a fixed **internet address / domain** (e.g. `https://vote.htl-wels.at`) instead of a changing PC IP.
- Voters no longer need to be on the same WiFi.
- No firewall configuration on an individual PC required.

B2. A real, trusted certificate

The production version uses an **official TLS certificate** (e.g. via Let's Encrypt) for the school domain. This removes:

- the manual installation of `votevox-rootCA.pem` on every phone,
- every certificate warning.

The camera works on all devices immediately.

B3. Multiple administrator accounts & password policy

- Personal admin accounts instead of one shared default login.
- Forced password change on first login, strong password policy.
- Optional: roles (e.g. "election manager" vs. "read/results only").
- Integration with the existing school sign-in (e.g. Microsoft 365 / LDAP).

B4. Security & operations

- Strong, randomly generated **JWT secret** and **DB password** (not the test defaults).
- Regular automatic **database backups**.
- Encrypted connections throughout, a hardened server operating system.
- Logging/monitoring for operations.

B5. Convenience features (possible extensions)

- **Per-class code distribution** as separate PDFs / batch printing.
 - A live turnout dashboard on a projector (without interim results).
 - Export of the results as a PDF report for the election committee.
 - Multilingual interface (DE/EN).
 - A status view of which classes have not yet fully voted (still anonymous, only at class level).
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Status: test program of the diploma project. The default credentials and test secrets are intended for the demo only and must be changed before any real use.